Term 3: Checklist

Topic	Learning Outcomes	RAG Yourself	
		1	2
Y7 Chemicals (Revision)	Describe what an element is in terms of types of atoms		
	Describe how a molecule is formed from atoms		
	Compare chemical compounds to mixtures		
	Explain why chemical compounds are hard to separate		
	Draw particle diagrams of the three states of matter: solids, liquids and gases		
	State the physical properties of the three states of matter		
	Compare the kinetic energy of particles in different states of matter		
	Explain why solids and liquids cannot be compressed easily, but gases can		
	Describe what happens to the particles in a substance during a change of state		
	Describe how litmus paper can be used to test acids and base		
	Describe how universal indicator can be used to test acids and bases		
	Match colours to numbers on the pH scale		
	Compare how useful different pH indicators are		
	Describe what a neutralisation reaction is		
Y7 Forces and Motion (Revision)	State a range of contact and non-contact forces		
	Draw force arrows on objects that are at rest or moving		
	Calculate the resultant force of an object when two forces are acting on it		
	Explain why an object will start to accelerate using the idea of forces		
	Explain why an object will travel at a steady speed using the idea of forces		
	State the units of mass and weight		
	Explain why your weight changes on different planets but your mass does not		
	Calculate your mass or weight. Try this one: Earth's gravity is 9.8N/kg, Mr Wormstone has a mass of 78kg. What is his weight?		
	Explain why friction occurs between two surfaces		
	Describe the difference between the independent and dependent and control variables		
Y8 Heating and Cooling (Revision)	Explain the properties of solids, liquids and gases in terms of the arrangement of particles e.g. why can gasses be compressed?		
	Describe what is meant by expansion and contraction		
	Describe what happens to the particles in a substance when it is heated		
	Describe how thermal energy transfers between hotter and colder objects		
	Give the standard units of temperature and energy		

	Describe how thermal energy transfer occurs through solids through conduction		
	Define the terms insulator and conductor with examples of each		
	Explain why metals are good conductors of energy		
	Describe how thermal energy transfers through liquids and gases by convection		
	Explain why the colour of a house is important when considering thermal insulation		
Periodic Table	Describe what an atom is and state their approximate size		
	Compare atoms and elements		
	Draw particle diagrams showing elements and compounds		
	Describe what a semi-metal is		
	Describe the general properties of metals and non-metals		
	Describe how the periodic table is arranged in groups and periods		
	Identify the group and period of an element on the periodic table		
	Define "subatomic particle"		
	Draw a labelled diagram of an atom including: nucleus, protons, neutrons, shells and electrons		
	State the charge of the three subatomic particles		
	State the names of groups 1, 2, 7 and 8 of the periodic table		
	Describe the properties of elements in group 7 (HT only)		
	Describe the trend in reactivity down group 7 (HT only)		
	Explain the trend in reactivity down group 7 in terms of electronic structure (HT only)		
Staying Alive	Describe the role of stem cells in the body		
	Compare asexual and sexual reproduction		
	Describe the advantages and disadvantages of asexual reproduction		
	Describe the steps involved in plant reproduction (pollination, fertilisation, germination)		
	Label the reproductive parts of a plant		
	Describe how fertilisation happens in animals		
	Compare gestation periods of animals to their size		
	Describe the main changes that occur during puberty		
	Describe how puberty is controlled by hormones		
	State the two main sex hormones and where they are produced		
	Label the male reproductive organs		
	Label the female reproductive organs		
	Describe the function of the male reproductive system		
	Describe the function of the female reproductive system		
	Describe the steps involved in the menstrual cycle		
	Describe the impact of the menopause on female fertility		